

SQL Joins Assignment

Table 1: Customers

CustomerID	CustomerName	City
1	John Smith	New York
2	Mary Johnson	Chicago
3	Peter Adams	Los Angeles
4	Nancy Miller	Houston
5	Robert White	Miami

Table 2: Orders

OrderID	CustomerID	OrderDate	Amount
101	1	2024-10-01	250
102	2	2024-10-05	300
103	1	2024-10-07	150
104	3	2024-10-10	450
105	6	2024-10-12	400

Table 3: Payments

PaymentID	CustomerID	PaymentDate	Amount
P001	1	2024-10-02	250
P002	2	2024-10-06	300
P003	3	2024-10-11	450
P004	4	2024-10-15	200

Table 4: Employees

EmployeeID	EmployeeName	ManagerID
1	Alex Green	NULL
2	Brian Lee	1
3	Carol Ray	1
4	David Kim	2
5	Eva Smith	2

Question 1. Retrieve all customers who have placed at least one order.

Answer : The SQL Query we run to Complete the given Task is:

Select Distinct Customer.Customer_ID, Customer.Customer_Name, Customer.City,
Orders.Amount, Orders.Order_ID, Orders.Order_Date From Customer Inner Join orders ON
Customer.Customer_ID = Orders.Customer_ID;

The Result Grid we Got:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Customer_ID	Customer_Name	City	Amount	Order_ID	Order_Date
▶	1	John Smith	New York	250	101	2024-10-01
	2	Mary Johnson	Chicago	300	102	2024-10-05
	1	John Smith	New York	150	103	2024-10-07
	3	Peter Adams	Los Angeles	450	104	2024-10-10

Question 2. Retrieve all customers and their orders, including customers who have not placed any orders.

Answer : The SQL Query we run to Complete the given Task is:

SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City,
Orders.Order_ID, Orders.Order_Date, Orders.Amount
FROM Customer
LEFT JOIN Orders ON Customer.Customer_ID = Orders.Customer_ID;

The Result Grid we Got:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Customer_ID	Customer_Name	City	Order_ID	Order_Date	Amount
▶	1	John Smith	New York	103	2024-10-07	150
	1	John Smith	New York	101	2024-10-01	250
	2	Mary Johnson	Chicago	102	2024-10-05	300
	3	Peter Adams	Los Angeles	104	2024-10-10	450
	4	Nancy Miller	Houston	NULL	NULL	NULL
	5	Robert White	Miami	NULL	NULL	NULL

Question 3. Retrieve all orders and their corresponding customers, including orders placed by unknown customers.

Answer : The SQL Query we run to Complete the given Task is:

Select Customer.Customer_ID, Customer.Customer_Name, Customer.City,

Orders.Order_ID, Orders.Order_Date, Orders.Amount

From Customer

Right Join Orders On Customer.Customer_ID = Orders.Customer_ID;

The Result Grid we Got:

Result Grid		Filter Rows:		Export:	Wrap Cell Content: IA	
	Customer_ID	Customer_Name	City	Order_ID	Order_Date	Amount
▶	1	John Smith	New York	101	2024-10-01	250
	2	Mary Johnson	Chicago	102	2024-10-05	300
	1	John Smith	New York	103	2024-10-07	150
	3	Peter Adams	Los Angeles	104	2024-10-10	450
	NULL	NULL	NULL	105	2024-10-12	400

Question 4. Display all customers and orders, whether matched or not.

Answer : The SQL Query we run to Complete the given Task is:

SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City,

Orders.Order_ID, Orders.Order_Date, Orders.Amount

FROM Customer

LEFT JOIN Orders ON Customer.Customer_ID = Orders.Customer_ID

UNION

SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City,

Orders.Order_ID, Orders.Order_Date, Orders.Amount

FROM Customer

RIGHT JOIN Orders ON Customer.Customer_ID = Orders.Customer_ID;

The Result Grid we Got:

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	Customer_ID	Customer_Name	City	Order_ID	Order_Date	Amount
▶	1	John Smith	New York	103	2024-10-07	150
	1	John Smith	New York	101	2024-10-01	250
	2	Mary Johnson	Chicago	102	2024-10-05	300
	3	Peter Adams	Los Angeles	104	2024-10-10	450
	4	Nancy Miller	Houston	NULL	NULL	NULL
	5	Robert White	Miami	NULL	NULL	NULL
	NULL	NULL	NULL	105	2024-10-12	400

Question 5. Find customers who have not placed any orders.

Answer : The SQL Query we run to Complete the given Task is:

```
SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City
FROM Customer
LEFT JOIN Orders ON Customer.Customer_ID = Orders.Customer_ID
WHERE Orders.Order_ID IS NULL;
```

The Result Grid we Got:

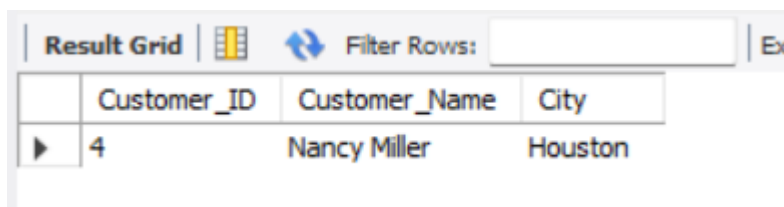
Result Grid	Filter Rows:	Export:
Customer_ID	Customer_Name	City
4	Nancy Miller	Houston
5	Robert White	Miami

Question 6. Retrieve customers who made payments but did not place any orders.

Answer : The SQL Query we run to Complete the given Task is:

```
SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City
FROM Customer
INNER JOIN Payments ON Customer.Customer_ID = Payments.Customer_ID
LEFT JOIN Orders ON Customer.Customer_ID = Orders.Customer_ID
WHERE Orders.Order_ID IS NULL;
```

The Result Grid we Got:



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with three columns: 'Customer_ID', 'Customer_Name', and 'City'. There is one row of data with the values 4, Nancy Miller, and Houston. Above the table, there is a 'Filter Rows:' input field and an 'Export' button.

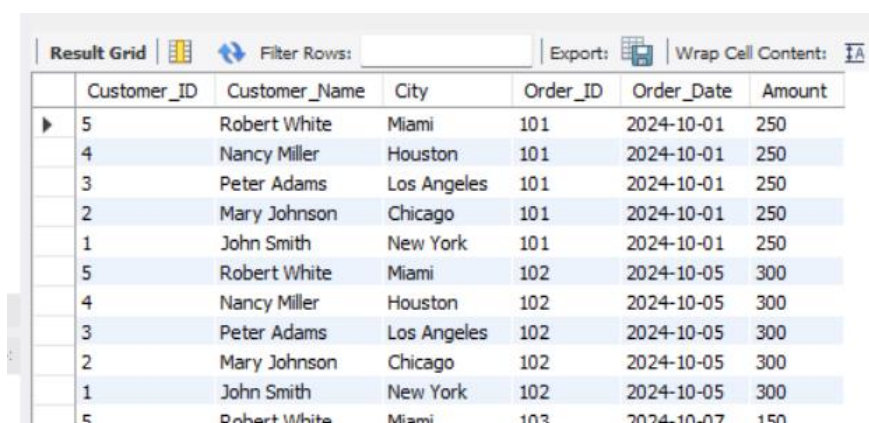
	Customer_ID	Customer_Name	City
▶	4	Nancy Miller	Houston

Question 7. Generate a list of all possible combinations between Customers and Orders.

Answer : The SQL Query we run to Complete the given Task is:

```
SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City,
Orders.Order_ID, Orders.Order_Date, Orders.Amount
FROM Customer
cross Join Orders;
```

The Result Grid we Got:



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with six columns: 'Customer_ID', 'Customer_Name', 'City', 'Order_ID', 'Order_Date', and 'Amount'. There are 15 rows of data, representing all combinations of 5 customers and 3 orders. The rows are numbered 1 to 15 on the left. Above the table, there is a 'Filter Rows:' input field, an 'Export' button, and a 'Wrap Cell Content:' checkbox.

	Customer_ID	Customer_Name	City	Order_ID	Order_Date	Amount
▶	5	Robert White	Miami	101	2024-10-01	250
	4	Nancy Miller	Houston	101	2024-10-01	250
	3	Peter Adams	Los Angeles	101	2024-10-01	250
	2	Mary Johnson	Chicago	101	2024-10-01	250
	1	John Smith	New York	101	2024-10-01	250
	5	Robert White	Miami	102	2024-10-05	300
	4	Nancy Miller	Houston	102	2024-10-05	300
	3	Peter Adams	Los Angeles	102	2024-10-05	300
	2	Mary Johnson	Chicago	102	2024-10-05	300
	1	John Smith	New York	102	2024-10-05	300
	5	Robert White	Miami	103	2024-10-07	150

Question 8. Show all customers along with order and payment amounts in one table.

Answer : The SQL Query we run to Complete the given Task is:

```
SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City,
Orders.Order_ID, Orders.Amount AS Order_Amount, Payments.Payment_ID,
Payments.Amount AS Payment_Amount FROM Customer LEFT JOIN Orders ON
Customer.Customer_ID = Orders.Customer_ID LEFT JOIN Payments ON
Customer.Customer_ID = Payments.Customer_ID;
```

The Result Grid we Got:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Customer_ID	Customer_Name	City	Order_ID	Order_Amount	Payment_ID	Payment_Amount
▶	1	John Smith	New York	103	150	P001	250
	1	John Smith	New York	101	250	P001	250
	2	Mary Johnson	Chicago	102	300	P002	300
	3	Peter Adams	Los Angeles	104	450	P003	450
	4	Nancy Miller	Houston	NULL	NULL	P004	200
	5	Robert White	Miami	NULL	NULL	NULL	NULL

Question 9. Retrieve all customers who have both placed orders and made payments

Answer : The SQL Query we run to Complete the given Task is:

```
SELECT Customer.Customer_ID, Customer.Customer_Name, Customer.City,
Orders.Order_ID, Orders.Amount, Payments.Payment_ID, Payments.Amount FROM
Customer INNER JOIN Orders ON Customer.Customer_ID = Orders.Customer_ID INNER
JOIN Payments ON Customer.Customer_ID = Payments.Customer_ID;
```

The Result Grid we Got:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Customer_ID	Customer_Name	City	Order_ID	Amount	Payment_ID	Amount
▶	1	John Smith	New York	101	250	P001	250
	2	Mary Johnson	Chicago	102	300	P002	300
	1	John Smith	New York	103	150	P001	250
	3	Peter Adams	Los Angeles	104	450	P003	450